

Generation Interconnection System Impact Study Report

For Queue Project Q20-1471

Drain 138 kV

July 01, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1471. The Interconnection Customer proposes a 74.5 MW battery energy storage facility interconnecting at the Drain 138 kV substation in the Virginia area. The request was received on May 04, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 524 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,348,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mohican – Charlottesville 138kV L	Red Hill 345-138 kV Transformer Bank	137.8	168	10.9
North Riv D.P – Downtown 138kV	Weyers Cave – Grottoes 138kV	127.8	209	61.0
Mohican – Barracks Road Substation 138kV	Red Hill Substation 345 kV/138 kV TX	122.8	200	10.9
Barracks Rd – Timken Mercy 138kV	Red Hill 345/138kV Transformer Bank	122.1	223	10.9
Declo – Salamancarocheater St 138kV L	Aspen – Crandall 345kV	120.6	287	16.6
Card – Yucipa 138kV line	Crandall – Aspen 345kV	117.9	267	8.7
Card – Craggy 138kV	Crandall – Aspen 345kV	116.8	258	8.7
Craggy – Stuarts Draft 138kV Line	Crandall – Aspen 345kV	113.9	239	8.7
Stuarts Draft – Harry Allen 138kV	Crandall – Aspen 345kV L	111.7	233	8.7
Erie – Fishersville 138kV	Crandall – Aspen 345kV	105.8	201	8.7
Fishersville – Harry Allen 138kV	Crandall – Aspen 345kV	105.7	219	8.7
Philpott – Williams Lake 138kV	Mountaineer Wind Energy Ctr – Santee Cooper 345kV	105.0	321	8.9
Philpott – Folsom 138kV L	Mounaineer Wind Energy Ctr – Santee Cooper 345kV	104.7	328	8.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Charlottesville – Mohican 138kV	Rebuild line with higher-capacity conductor	\$1,890,000	14% (\$264,000)
North Riv D.P – Downtown 138kV	Rebuild line with higher-capacity conductor	\$4,086,000	57% (\$2,333,000)
Barracks Rd – Mohican 138kV line	Rebuild line with higher-capacity conductor	\$7,030,000	14% (\$981,000)
Barracks Road Substation – Timken Mercy 138kV line	Rebuild line with higher-capacity conductor	\$8,200,000	14% (\$1,144,000)
Declo – Salamancarocheater St 138kV L	Rebuild line with higher-capacity conductor	\$40,824,000	16% (\$6,397,000)
Card – Yucipa 138kV line	Rebuild line with higher-capacity conductor	\$4,683,000	10% (\$465,000)
Card – Craggy 138kV	Rebuild line with higher-capacity conductor	\$9,670,000	10% (\$959,000)
Craggy – Stuarts Draft 138kV L	Rebuild line with higher-capacity conductor	\$5,944,000	10% (\$590,000)
Stuarts Draft – Harry Allen 138kV	Rebuild line with higher-capacity conductor	\$5,265,000	10% (\$522,000)
Erie – Fishersville 138kV	Reconductor line	\$1,273,000	10% (\$126,000)
Fishersville – Harry Allen 138kV	Reconductor line	\$937,000	10% (\$93,000)
Philpott – Williams Lake 138kV	Reconductor line	\$2,725,000	7% (\$197,000)
Philpott – Folsom 138kV L	Reconductor line	\$3,829,000	7% (\$277,000)
	Total allocated to project		\$14,348,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.